

Leaving groups: Please see the other document on a few rules for Organic Chemistry on bases and nucleophiles.

Comparing leaving groups:

It is important to remember that this is a complex topic. THEREFORE, we need to simplify for this course. To keep it simple we will compare leaving groups on systems which are VERY similar with only one structural variable different: FOR example:  $\text{H}_2\text{O}$  vs  $\text{OH}^-$  (everything the same, but one is charged, the other isn't). OR  $\text{Cl}^-$  vs  $\text{Br}^-$  (same valence electrons, but different period). OR  $\text{NH}_3$  vs  $\text{H}_2\text{O}$  (same period, same atom attached to central atom, same charge, but different family or periodic table group). It isn't easy (therefore we will not do it) to compare leaving groups with more than two structural changes: for example  $\text{OH}^-$  vs  $\text{NH}_3$ .

Because leaving groups are related to nucleophiles/bases, please remember acid/base Bronsted theory (each acid will have a conjugate base, and viceversa): the stronger the acid, the weaker the conjugate base (the opposite is also true):

SO, HCl is a strong acid ( $\text{Cl}^-$  is a weak base).  $\text{H}_2\text{O}$  is a weak acid ( $\text{OH}^-$  is a strong base).

Normally, the stronger the base/nucleophile, the worse the leaving group. (See exceptions below).

So, for example:

-If everything else is the same, the negatively charged species is always a MUCH stronger base and nucleophile, therefore a worse leaving group:

**$\text{OH}^-$  worse LG than  $\text{H}_2\text{O}$**

-If everything else is the same, going left in the same period of the periodic table increases the strength of the base AND nucleophile, therefore the worse the leaving group:

**$\text{CH}_3^-$  worse LG than  $\text{NH}_2^-$  worse LG than  $\text{OH}^-$  (we stop at O in the first period because F is "special");  $\text{NH}_3$  worse LG than  $\text{H}_2\text{O}$  ; etc**

This means: Alkyl groups are NOT leaving groups (at all), and N-centred groups such as  $\text{NH}_2$ ,  $\text{NR}_2$  or similar, are NOT either. F should NOT be considered a leaving group (the chemistry of F is very weird, very different to anything else, just remember the C-F is too strong to break through normal pathways). So, the summary of these previous rules are:  **$\text{OH}^-$  is a poor leaving group,  $\text{H}_2\text{O}$  is a good leaving group. We will always attempt to convert an  $\text{R-OH}$  into an  $\text{R-OH}_2^+$  if we want  $-\text{OH}$  to leave a molecule.**

**Main exception to the rule "the stronger the base, the stronger the nucleophile, therefore the worse leaving group":** If everything else is the same, going down in the same group increases the strength of the nucleophile, BUT, decreases the strength of the base, and increases the better leaving group: SO:

**Base trend:**  $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$  ; **Nucleophile trend:**  $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$ ;

**Leaving group trend:**  $\text{F}^-$  worse LG than  $\text{Cl}^-$  worse LG than  $\text{Br}^-$  worse LG than  $\text{I}^-$

( $\text{F}^-$  is NOT a LG,  $\text{Cl}^-$  is an OK leaving group,  $\text{Br}^-$  very good LG,  $\text{I}^-$  great LG).

None of the other groups we considered for base/nucleophile trends in the previous document are considered good LG with the exception of  $\text{H}_2\text{O}$ .

**If you remember only a few examples, please remember:**

Good/OK nucleophiles but weak base, and OK to very good leaving groups:  $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{I}^-$

Good nucleophile, good base, very poor leaving group:  $\text{OH}^-$

Poor nucleophile, poor base, good leaving group:  $\text{H}_2\text{O}$

$\text{F}^-$ ,  $\text{RO}^-$ ,  $\text{ROH}$ ,  $\text{NH}_2^-$ ,  $\text{NH}_3$ ,  $\text{NR}_2^-$ ,  $\text{NHR}_2$ ,  $\text{NR}_3$  are NOT Leaving groups at all in  $\text{S}_\text{N}1/\text{S}_\text{N}2/\text{E}1/\text{E}2$  (except for some very special reactions that we have NOT seen in class, therefore not considered in this course: I am just warning you in case you see these in a book or online)

$\text{ROH}/\text{RO}^-/\text{NH}_2/\text{NH}_3/\text{NR}_2/\text{NHR}_2$  are only leaving groups in hydrolysis of esters or amides (the negatively charged versions are leaving groups in basic hydrolysis esters/amides and the neutral ones in the acid hydrolysis of esters/amides).